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MSc Artificial Intelligence

Learning Interpretable Feature Spaces
via Domain-Informed Encoders

MSc Dissertation — Module CS4700A

Submitted by:

Bhoopesh Nandan Singh
Student Id: 250142934
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Supervisor : **Dr. Roberto Alamino**

This dissertation is submitted in partial fulfillment of the requirements for the award of MSc Artificial Intelligence at Aston University.

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Abstract

In most concept-driven interpretability approaches, the process starts from an already defined set. The expert or scientist decides beforehand on which concepts are important and marks examples of such concepts. However, this is not possible when interpreting data in the clinic, since the concepts needed are the ones to be uncovered.

This dissertation introduces the Domain-Informed Feature Encoder (DIFE) pipeline, which contradicts this assumption by turning it around. This approach extends DINO Vision Transformers for image data and employs our own PanelFTTransformer for clinical blood panels. Potential concepts are detected using Gaussian Mixture Model clustering, followed by assigning names to them using a human evaluator or LLM in charge of the task – labeling happens post-training, and no concept labels are utilized throughout the whole process of training.

As for the performance in the image domain, it greatly surpassed all expectations. Multi-level feature extraction from blocks 8, 10, and 12 provided a 75% increase in silhouette scores compared to single-level feature extraction. Moreover, a concept-based classifier reached 98.9% accuracy on the test dataset and scored a macro F1 of 0.99 – this result significantly exceeds the accuracy obtained using the raw feature baseline. As for the clinical tabular data domain, patterns corresponding to dyslipidemia, metabolic syndrome, and kidney damage were extracted from the data containing information about 18,673 patients. A classifier based on three classes reached 50.9% accuracy, which is considerably higher than 33.3%, which is the baseline of a random classifier.

One consistent trend emerged and proved itself again: the interpretable classifier outperformed the baseline. To be exact, contrary to previous expectations, there were no trade-offs with regards to the accuracy.

Keywords: interpretable machine learning, self-supervised learning, Vision Transformer, DINO, FT-Transformer, Gaussian Mixture Model, concept activation vectors, unsupervised concept discovery, panel positional encoding, clinical AI

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1. Introduction

1.1 Motivation

Models like Vision Transformers are good at making predictions but poor at explaining how they were made. VT models can classify chest X-rays to radiologist accuracy level, while FT-Transformers can detect diabetic risk in patients' blood panels with reasonable precision. Once asked to explain why exactly they reached their conclusions, they produce a 384-dimensional vector with little relevance to anything that could be clinically meaningful to humans. This gap between prediction capability and explainability is what this dissertation tackles. While post hoc explanation techniques, such as GradCAM, may be able to highlight which pixels are important to the final decision, attributions and concept-level explanations are not the same thing. Even if there is some alignment between attribution maps and the actual lesion's border, this doesn't mean that the model has considered lesion boundary irregularity in making a decision, something that could be clinically useful, whereas pixel attributions imply extra mental effort on the part of the practitioner. Concept-based methods like TCAV (Kim et al., 2018) and Concept Bottleneck Models (CBM; Koh et al., 2020) attempt to bridge the gap between machine and man through meaningful concept extraction. Yet they too rely on predetermined concept labels before training, a necessary prerequisite that may not exist in real-world applications, particularly clinical ones. The proposed solution embeds some information about domain structure (panel composition in the case of tabular data, spatial parts in the case of vision tasks) into the architecture of the encoder and then extracts unlabeled concepts based on features. Provided the architecture is sound, latent-space clusters will represent concepts as would be identified by domain experts. Experiments included in this dissertation demonstrate that this is true at least in the two domains analyzed.

1.2 The Three Literature Gaps

There are at least three limitations found in the literature reviewed to motivate the research conducted in this thesis. The aforementioned limitations will be elaborated in Chapter 2 below.

Gap 1 – the need to know the concepts in advance. TCAV and CBMs require some concept samples before training, even though ACE (Ghorbani et al., 2019) automates the candidate generation process. Hence, these methods are confined to only certain domains where the relevant concepts have already been determined.

Gap 2 – underexploration of intermediate transformer layers. According to Caron et al. (2021), the architecture of the intermediate layers of the DINO framework has been intentionally overlooked, a conclusion echoed for tabular transformers in Gorishniy et al. (2021). Thus, intermediate representations are treated as mere implementation details.

Gap 3 – lack of an LLM-driven concept naming mechanism for clinical numerical data. ClusterLLM (Zhang et al., 2023) proves that it is possible for an LLM to help name clusters discovered in the domain of natural language processing. Yet no method has ever applied an LLM for concept naming in the numerical clinical domain, let alone verifying its clinical sense.

1.3 The Three Design Pillars

The first design choice follows from Gap 1. Namely, concepts can be discovered using a Gaussian Mixture Model clustering of feature vectors generated by the encoder. Labelling, whether performed by a human or the LLM, occurs after concept discovery, not before.

The second design choice addresses Gap 2: features should be extracted from multiple layers of intermediate transformer blocks and concatenated. For DINO this means layers 8, 10, and 12; for PSFTT, this implies layers L2, L4, and L6. Additionally, a foreground mask guided by the attention maps is used for image features. For tabular clinical features, an additional source of domain structure information is provided through panel positional encoding and two-scale supervised self-learning. Based on the taxonomy suggested by Von Rueden et al. (2021), the design principles are consistent with the following three categories of domain knowledge embedding: architectural embedding (multiple layer extraction, foreground masking, panel position embedding), loss-based objectives (fine-tuning semantic consistency, two-level self-supervised learning), and input embedding (clinical deviation normalization).

The third design choice arises from Gap 3: the discovered concepts should receive semantic labels after concept discovery. Image clusters receive labels using a Streamlit interface showing sample patches. Tabular clusters receive labels using an LLM Clinical Advisor assessing a patient profile deviation and providing a structured clinical diagnosis, which is then evaluated and, if needed, corrected by a human.

For each design principle, an ablation experiment is performed to establish its role. The cause-and-effect relationship between the gap, design decision, and experiment forms the basis of the result presentation structure.

1.4 Two-Domain Validation

The methodology is validated in two domains – visual (image data) and clinical (tabular data). The former is chosen to enable visual inspection of concept quality, while the latter represents the real-life clinical application scenario for which the pipeline was conceived. The same downstream process of clustering, labelling, concept evaluation, and classification is implemented in both domains without domain-specific stages of implementation to prove domain generality (NFR5).

A visual dataset allows a proof-of-concept demonstration, where the quality of discovered concepts is evident through human inspection of how well a named concept matches its associated visual patches—for instance, inspecting whether "cat ear" indeed looks like the part of an image associated with that name. This is impossible in the case of a medical tabular dataset, because in the latter the meaning of a cluster cannot be understood visually but only by analyzing its deviation profile. By starting with a proof-of-concept in the image domain, one establishes a verifiable benchmark before extending the architecture into the more challenging clinical setting.

On the other hand, the tabular domain, based on NHANES blood panel data, provides the second, co-equal role: it checks whether the concept discovery framework is indeed generally applicable to an entirely different data modality—one which lacks any spatial structure, features heterogeneous scale, and is characterized by explicitly clinically defined normal ranges. Should this framework be able to yield only meaningful concepts in the image domain, the proposed solution would not go beyond computer vision applications. The fact that it can produce meaningful clinical concepts without any outcome labels marks this work as a domain-general interpretability framework.

Taken together, both domains make up an ideal validation pair: a human-verifiable semantic validation for images and a clinically meaningful validation for tables. In no previously explored concept-discovery framework discussed in Chapter 2 did these two aspects coincide.

1.5 Aim and Objectives

Aim: Design, implement, and evaluate a pipeline capable of discovering interpretable concepts from unlabeled data, labeling them semantically using domain knowledge, and enabling predictions grounded in the named concepts using image and clinical tabular data.

Objectives:

Objective 1. Create a pipeline for concept discovery involving foreground masking, spatial data augmentation, and semantic fine-tuning of the DINO ViT-S/8 model, with the caveat that there will be no labeled data involved in any step of the process. Deliverable: Python code repository with the implementation on GitHub. Due date: End of Sprint 3.

Objective 2. Perform ablation studies comparing different approaches to feature extraction and analyze their performance using metrics such as silhouette score, normalized mutual information (NMI), class purity, and AUC-ROC. Deliverable: Table with the results of ablation studies. Due date: End of Sprint 5.

Objective 3. Develop a pipeline for the parallel analysis of NHANES blood panel data using PanelFTTransformer encoder and concept labeling driven by large language models. Deliverable: Results obtained through the labeling and scoring pipelines along with clinical concept table. Due date: End of Sprint 6.

Objective 4. Confirm that the performance of the concept-based classifier is comparable to that of the feature-based classifier. Deliverable: Table summarizing the classification results. Due date: End of Sprint 7.

1.6 Legal, Ethical and Professional Context

Both datasets are publicly available and anonymized. STL-10 can be freely used for academic purposes (Coates et al., 2011). NHANES datasets are published by the CDC and all the identifying variables are stripped. Annotations for Oxford-IIIT pet mask (Parkhi et al., 2012) are only used on 100-image validation set to measure foreground mask quality, not for training. Weights of the DINO model are provided under the Apache 2.0 license. Labeling tools operate exclusively in local mode and do not transmit data anywhere. The clinical application motivation aligns with GDPR Article 22 and EU AI Act. The explanation format ("Prediction: Metabolic syndrome (87% confidence); driven by elevated glucose pattern (0.71) and borderline lipid pattern (0.43)") complies with the essence of the Right to Explanation. The human-in-the-loop component of the process conforms to the EU AI Act high-risk category.

1.7 Dissertation Structure

Chapter 2 reviews the related literature within four domains and ties each one to the one of the three gaps. Chapter 3 provides a detailed list of the formal requirements. Chapter 4 describes design and rationales. Chapter 5 deals with project management. Chapter 6 presents experimental results and reflections. Chapter 7 provides concluding remarks.

2. Literature Review and Contextual Analysis

In this section, the literature review gaps that are used to justify each design choice will be outlined in relation to the three literature gaps stated in section 1.2. Following the discussion of each gap, the domain context will also be outlined.

2.1 Concept-Based Interpretability - Gap 1

The key problem posed by black-box algorithms in clinical practice is not related to the quality of predictions, which is generally high in many cases. The core problem lies in the inability to question the reason behind any particular prediction. TCAV (Kim et al., 2018) provides an effective solution for this problem. The idea of this approach involves training a linear probe that establishes connections between activation vectors in a layer and a binary indicator of concept presence. Subsequently, this vector is used to evaluate how sensitive the model is to the corresponding concept. The critical drawback of this technique, mentioned by the authors, is the need for hundreds of manually labeled examples per each concept, which in domains such as medical imaging, where the list of possible concepts for a novel disease presentation is not known beforehand, poses an insurmountable constraint.

Concept Bottleneck Models (Koh et al., 2020) solve the above problem by enforcing predictions to pass through a bottleneck layer with encoded concept activations. This approach allows for achieving great results with plenty of concept supervision; however, without it, a CBM becomes equivalent to a black-box model, because the bottleneck layer learns arbitrary activations and ignores any semantics.

Unlike TCAV, ACE (Ghorbani et al., 2019) does not require a pre-labeled data set to generate per-concept explanations because it uses an already trained classification model to extract concept candidates automatically. The approach involves cropping the input image into superpixels and evaluating each of them according to the TCAV technique. Still, ACE depends on segmentation: it needs an additional algorithm to come up with the crop suggestions. Moreover, this problem is specifically highlighted by the authors as one to address in future work. In the present case study, this gap is addressed by leveraging the model's own attention weights to generate the patches.

While STEGO (Hamilton et al., 2022) proves that DINO's representation is rich enough to allow for unsupervised segmentation into semantically coherent regions, it stops here without

providing a naming pipeline, a concept-based classifier, and per-instance explanations. The latter problem, together with naming, makes up the core of the current research.

2.2 Self-Supervised Visual Encoders - Gap 2 (Image Domain)

Contrastive self-supervised techniques, exemplified by SimCLR (Chen et al., 2020) and MoCo (He et al., 2020), provided a breakthrough in learning visual representation without ground truth labels. Such algorithms were intended for global image-level representations without any specific objectives for spatial part representations.

The next milestone was reached by DINO (Caron et al., 2021), who have reported surprising results: self-distillation objective yielded attention maps for [CLS] tokens segmenting foreground objects, as well as patch features clustering according to semantics without any part-based supervision. The authors report this fact and state that the structure of layers is "left unexplored." In some sense, the current research project may be viewed as an embodiment of this gesture made but left unexplored by DINO's authors.

Instead of feature extraction in the last block, as Amir et al. (2022) do in their ViT-based part discovery task, the features will be extracted in Blocks 8, 10, and 12 and concatenated into a single vector of size 1,152. It will be assumed that Block 8 contains features encoding textures, Block 10 encodes part outline features, while Block 12 encodes semantics of part identity. The corresponding ablation variant (4.5) will test whether this assumption holds true.

The foreground masking procedure in DIFE will follow the LEOPART pipeline (Ziegler and Asano, 2022), using [CLS] attention density for DINO to separate foreground objects from the background before performing clustering. Specifically, the threshold $\theta = 0.75$ will be used, selected via a grid search against Oxford-IIIT Pet Dataset annotations (mean IoU 0.73 ± 0.09). At the same time, Siméoni et al.'s LOST (2021) independently confirms that ViT-based patch features support localization without supervision.

An interesting twist to this technique is that [CLS] attention is used here only as a binary indicator, selecting foreground patches, not as an explanation of the importance of features inside it. It resolves the controversy over interpretation of attention weights (see Jain and Wallace, 2019; Wiegreffe and Pinter, 2019).

2.3 Tabular Deep Learning and Panel Structure - Gap 2 (Tabular Domain)

The FT-Transformer model (Gorishniy et al., 2021) continues to dominate tabular deep learning. The key idea behind it—the use of individual linear projections specific to a task, rather than shared embeddings, to represent each of the features—finds strong justification, since different physiological states correspond to different deviations, e.g., 2σ shift in glucose vs. 2σ shift in creatinine, and therefore, shared embeddings make little sense here. At the same time, the FT-Transformer does not take into account the structure of the dataset or clinical interdependencies between features, which is duly recognized by its developers.

Lab-MAE (Restrepo et al., 2025) utilizes masked autoencoding for tabular data, following the correct overall strategy for self-supervised pre-training on laboratory panels. Lab-MAE suffers, however, from one drawback associated with a purely random masking procedure: Lab-MAE masks individual values regardless of their panel affiliation, which overlooks the inherent structure of blood tests (CBC, BMP, Lipid), corresponding to the physiology of distinct physiological systems.

The proposed PanelFTTransformer model (PSFTT) is based on the FT-Transformer framework with panel membership encoded explicitly and a self-supervised training scheme exploiting this panel structure. PSFTT uses two levels of self-supervised pre-training, and intermediate representation layers (L2, L4, and L6) are used for concept clustering—apparently for the first time in case of FT-Transformers.

The basic problem faced by both FT-Transformer, Lab-MAE, and related architectures is that they consider the tabular input as a single sequence of feature values, without taking into account the fact that laboratory tests have a natural organization in panels. The result of a CBC test can be considered a panel since the values contained in it correlate in a specific way, and an example of that is anaemia, which appears through correlated values of haemoglobin, MCV, and RDW. Using typical feature tokenization and random masking leads to a representation that is statistically valid but does not contain any clinically meaningful interpretation. This problem precisely represents what PSFTT addresses through the use of a panel positional embedding and two scales for SSL.

2.4 LLM-Mediated Concept Validation — Gap 3

A concise but valuable work is ClusterLLM (Zhang et al., 2023). The idea behind ClusterLLM can be easily summarized: LLMs have enough world knowledge to determine if

a group of text items is coherent semantically. The model showed good performance on textual data. It should be noted that Zhang et al. explicitly indicate numerical and clinical domains as those not covered at the moment: “extending to non-textual domains where LLM input formatting is non-trivial remains an open challenge.”

The use of the LLM in the context of this particular project tackles precisely this issue. Instead of providing raw numeric values to the LLM, which would confuse it, a deviation profile for every cluster is calculated. In other words, the results of each laboratory test are classified as very elevated, slightly elevated, normal, slightly lowered, or very lowered in relation to the clinical norm. This deviation profile is rephrased into a structured clinical question for the LLM, which generates a proposed label, differential diagnoses, and a confidence score. After that, the suggestions are evaluated, amended, or rejected by the human domain expert.

Providing numerical data as a deviation profile seems to be the key point of the whole approach. Initially, attempts to provide the numbers in the raw format led to non-specific and evasive answers from the LLM. However, a carefully crafted structure, ordered according to the degree of deviation and explicitly including the normal range, yielded much more relevant and concrete recommendations. Even though the creation of prompts involved more work than expected, it paid off.

2.5 Domain Knowledge Integration Taxonomy

According to Von Rueden et al. (2021), domain knowledge integration is said to operate on three levels: input (feature engineering), architecture (inductive biases), and loss (auxiliary objectives). Similarly, Dash et al. (2022) arrive at the same taxonomy, further proving with evidence from various domains that the combined effect of all three levels of domain knowledge consistently surpasses the effects achieved by any of them individually. This framework is also included in our DIFE pipeline through the use of all three components. In terms of inputs, clinical deviation encoding of tabular data is used to make sure that scales are calibrated not only in terms of statistics but clinical importance as well. In terms of architecture, foreground attention mask for image inputs and panel position embedding for tabular data are used. Losses include semantic consistency, fine-tuning of image inputs and two-scale self-supervised learning of tabular data.

2.6 Critical Gap Analysis

Table 1 below positions DIFE against related work across the dimensions that matter for this project.

Method	Supervision	Part Discovery	Named Concepts	Per-Image Explanation	Gaps Addressed
TCAV (Kim et al., 2018)	Supervised + concept labels	Partial	Yes (manual)	No	None — introduces Gap 1
CBM (Koh et al., 2020)	Supervised + concept labels	Yes	Yes (manual)	Yes	None — introduces Gap 1
ACE (Ghorbani et al., 2019)	Supervised + segmentation	Yes	Partial	No	None — introduces Gap 1
STEGO (Hamilton et al., 2022)	Self-supervised	Yes	No	No	None — no naming pipeline
Amir et al. (2022)	Self-supervised	Yes (single layer)	No	No	Partial Gap 2
ClusterLLM (Zhang et al., 2023)	Self-supervised	Text only	Yes (text)	No	None — introduces Gap 3
FT-Transformer (Gorishniy, 2021)	Supervised (tabular)	No	No	No	None — introduces Gap 2 (tabular)
DIFE — This work	Self-supervised only	Yes (multi-layer)	Yes (post-hoc)	Yes	Gaps 1, 2, 3 — all three

Table 1: Gap analysis — position of DIFE against related work

DIFE is the only approach among the comparative studies discussed that requires no labels to be trained, generates named concepts, offers instance-specific explanations, and can handle images and tables. It is this combination that forms the core of the argument presented in this dissertation.

Three key insights come out of Table 1 and need explicit clarification. The first insight is that no current technique simultaneously solves the three issues: each solution currently exists that solves one or two gaps but leaves others unsolved. TCAV and CBM solve concept-level interpretability, but they rely on labeling. ACE solves the labeling problem but generates anonymous clusters. Lastly, DINO and FT-Transformer generate rich representations, but concept structures were never studied within the intermediate layers of these transformers. DIFE becomes the first solution pipeline to combine these strengths without relying on concept supervision.

The second insight stems from the fact that the table's column dedicated to the tabular data field is by far the emptiest. In truth, literature bias exists here as well as research in the area of interpretable machine learning is largely focused on image and textual data, failing to consider clinical numerical data adequately.

Finally, instance-level explanation - identifying which concepts influenced the decision on a patient's record - is a capability not offered by any of the techniques discussed except for DIFE.

3. Problem Description and Requirements

3.1 Problem Statement

Problem statement: Given an input domain where interpretability of predictions is required, propose a system that (1) detects recurrent semantic patterns without the help of labeled concepts, (2) facilitates a process by which the domain expert can name these patterns after training in a lightweight post-hoc step, and (3) uses these named patterns for prediction in a way that can be traced, audited, and contested by the expert.

The following failure modes define the space of designs. The first failure mode is labeling-at-training-time, where TCAV and CBMs need concept labels before or during training; however, if we do not know the concepts or it is costly to get annotations before training, it will not work. The second failure mode is the naming problem, where methods using DINO detect semantically consistent clusters but only label them with an integer like "cluster 7," which is meaningless to a medical practitioner. The design proposal above avoids both failure modes.

The problem described in this paper is related to CS4700 Project Specification "*04_Interpretable_Space.pdf*," as provided by Dr. Roberto Alamino.

3.2 Functional Requirements

FR#	Requirement	Description and Acceptance Criterion
FR1	Multi-layer DINO Feature Extraction	Extract patch features from ViT-S/8 blocks 8, 10, 12; concatenate to 1152-dim. AC: output shape (N_images × 784 × 1152); assertion verified on first run.
FR2	Attention-Guided Foreground Masking	Threshold CLS-to-patch attention at θ=0.75 to retain top 25% of patches. AC: mean IoU vs Oxford-IIIT Pet segmentation mask ≥ 0.70 on 100-image sample.
FR3	PCA + Spatial Augmentation	Reduce 1152→128 dims via PCA; append weighted (row/28, col/28) coordinates. AC: cumulative explained variance ≥ 85%; augmented dim = 130.
FR4	K-Selection Diagnostics	`Compute WCSS and silhouette for K ∈ [K_min, K_max]; recommend argmax(silhouette). AC: diagnostic plots saved; recommended K returned as scalar.`
FR5	GMM Concept Clustering	Fit diagonal-covariance GMM with K components; produce hard and soft assignments. AC: convergence within 200 EM iterations; assignments cover all foreground patches.
FR6	Semantic Consistency Fine-Tuning	`Fine-tune last 2 DINO blocks using GMM pseudo-labels as contrastive pairs; no ground-truth labels. AC: post-FT silhouette ≥ pre-FT silhouette on validation set.`
FR7	Human Concept Labelling UI (Image)	`Streamlit interface showing ≥16 representative patches per cluster; accepts label, confidence, include/exclude. AC: labels.json written correctly.`
FR8	PanelFTTransformer Architecture	FeatureTokenizer (T_i = x_i * W_i + b_i) + panel positional encoding (learned nn.Embedding for 3 panels) + 6 TransformerEncoder layers. AC: panel-patch output shape (N_records × 3 × 384).

FR#	Requirement	Description and Acceptance Criterion
FR9	Two-Scale SSL Pre-Training (Lab)	Within-panel masking (40% of tests in one panel) + cross-panel dropout (drop entire panel; CLS predicts panel mean). AC: pre-training loss converges; NMI improves vs. no pre-training.
FR10	LLM Clinical Concept Advisor	Build deviation profile per cluster; query LLM for clinical label + rationale; present to human for validation. AC: acceptance rate logged; ≥70% accepted without modification.
FR11	Concept Activation Scoring	Compute cluster-proportion score per image/record. AC: score matrix shape (N × K); scores sum to ≤1.0.
FR12	Concept-Based Classification	Train OneClassSVM (image) or LogReg (tabular) on concept score vectors; evaluate AUC-ROC, F1. AC: results written to metrics.json; compared to raw-feature baseline.
FR13	Per-Instance Explanation	Generate 3-panel figure: original image | semantic concept map | concept contribution bar chart. AC: generated in ≤5 seconds on CPU.

Table 2: Functional Requirements Summary

3.3 Non-Functional Requirements

NFR#	Category	Requirement
NFR1	Performance	Image feature extraction: ≤30 min for 1,200 images on Apple MPS or CUDA; ≤90 min on CPU. Lab pre-training: runnable on CPU in ≤4 hours at reduced epoch count.
NFR2	Reproducibility	All results reproducible from the same random_seed in config.yaml. No randomised operations outside seeded contexts.
NFR3	Portability	Runs on macOS (MPS), Linux (CUDA), Windows (CPU) with no code changes beyond the device field in config.yaml.
NFR4	Modularity	Each pipeline stage independently re-executable without re-running prior stages, via file-system cache.
NFR5	Domain Generality	The same downstream pipeline (clustering, labelling, scoring, classification) must execute identically on both image and tabular encoders with no stage-specific code.
NFR6	Explainability	All model decisions traceable through the concept activation vector framework. No black-box components after feature extraction.
NFR7	Privacy	No personal data processed. No external API calls for data. All computation local.

Table 3: Non-Functional Requirements

4. Practical Realisation: Design, Implementation and Testing

This chapter describes the development and reasoning behind the work. Each design choice is associated with one of the three gaps discovered in the literature. Section 4.2 and Section 4.3 cover the image and tabular pipelines, respectively; Section 4.4 covers the shared downstream modules; and Section 4.5 covers the experimental design.

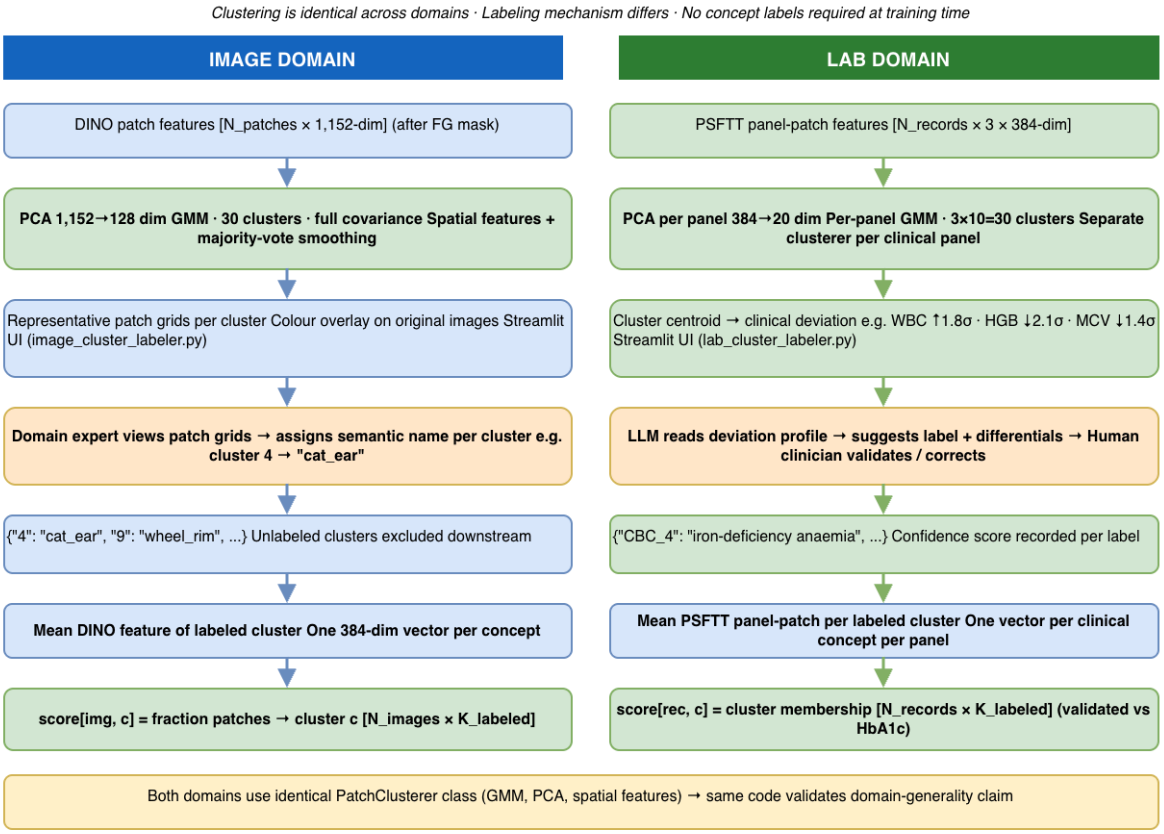

Figure 3: DIFE pipeline — nine stages applied identically across image and tabular domains.

4.1 Shared Architecture Overview

Both pipelines share a common approach with three main components: domain-aware embedding, clustering, and semantic annotation. The downstream pipeline of clustering, labeling, ranking of concepts, and classification uses the same software in both cases. The purposeful choice is made to be able to test NFR5 (domain generality) empirically rather than just claim that the requirement is satisfied.

Stage	Image Domain (DINO)	Tabular Domain (PSFTT)	Pillar
Encode	DINO ViT-S/8, blocks 8/10/12 → 1152-dim patch features	PSFTT: FeatureTokenizer + panel PE, layers L2/L4/L6 → 384-dim panel-patches	2
Filter	CLS attention masking: retain top 25% patches (θ=0.75)	Clinical deviation encoding: normalise to signed deviation from normal range	2

Stage	Image Domain (DINO)	Tabular Domain (PSFTT)	Pillar
Cluster	GMM (30 components, diag covariance) on PCA(128) + spatial(2) = 130-dim	Per-panel GMM (10 components × 3 panels = 30) on PCA(20) features	1
Label	Human expert names clusters via Streamlit UI (label_tool.py)	LLM Clinical Advisor suggests name; human validates (lab_label_tool.py)	3
Score	Cluster-proportion score: fraction of patches per concept per image	Soft GMM membership probability per concept per patient record	1
Classify	LogReg on concept scores; explain via 3-panel figure	LogReg on concept scores; compare to raw-feature LogReg baseline	1+2+3

Table 4: Two-Domain Architecture — same three pillars applied across both modalities

4.2 Image Domain — DINO Pipeline

4.2.1 Dataset and Preprocessing

The STL-10 dataset (Coates et al., 2011) consists of three classes, namely cat, car, and bird, with about 1,300 samples each. Cats form the positive class in this one-class classification model, similar to the disease-positive state in medicine. The quality of foreground masking was evaluated using Oxford-IIIT Pet segmentation masks (Parkhi et al., 2012) on 100 cat images that were not used in the training phase. The preprocessing process consisted of the following steps: resizing to 256, center cropping to 224, transformation to tensor format, and normalization using the ImageNet mean and standard deviation.

4.2.2 Multi-Layer Feature Extraction (Pillar 2 — Gap 2, Architecture)

The DINO ViT-S/8 model uses a 224×224 image which is divided into 784 patch tokens and passed through 12 layers. The model does not just use the output layer as the source of features but extracts features from three intermediate layers, features from three intermediate layers are concatenated as follows:

```
feats = torch.cat([all_layers[i][:, 1:, :] for i in (0, 2, 4)],
                                                          dim=-1)
# indices 0,2,4 of last-5 = blocks 7,9,11 = layers 8,10,12
```

Each patch feature vector will be 1152 dimensional. Block 8 encodes the mid-level texture structures, Block 10 encodes part outline information, whereas Block 12 encodes the part identity like “this is the cat’s ear.” In variant B I intend to check if using the concatenation of features results in better clustering of concepts than using only Block 12.

4.2.3 Attention-Guided Foreground Masking (Pillar 2 — Gap 1, Architecture)

The ACE paper by Ghorbani et al. (2019) highlights the elimination of dependency on pre-defined segmentation as potential research directions. In this study, foreground masks are

generated using the attention scores of the inherent [CLS] token from the DINO model across all six heads of the final transformer layer:

```
attn = model.get_last_selfattention(image)   # (1, 6, 785, 785)
cls_attn = attn[0, :, 0, 1:].mean(0)        # mean over heads ->
(784,)
mask = cls_attn > threshold(cls_attn, 0.75)  # retain top 25%
```

θ=0.75 was selected by grid search over {0.5, 0.6, 0.7, 0.75, 0.8} against Oxford-IIIT Pet segmentation mask annotations, achieving mean IoU 0.73 (±0.09) — the highest among all tested values.

4.2.4 GMM Concept Clustering (Pillar 1 — Gap 1)

After applying PCA to reduce the number of dimensions from 1152 to 128, while retaining 85.3 percent of variance and after adding the spatial dimension, which was surprisingly more effective than expected for preventing texture-based fusion, we fit the data into the Gaussian Mixture Model with Diagonal Covariance Matrix.

```
gmm = GaussianMixture(n_components=30, covariance_type='diag'
                                      ,n_init =3)
gmm.fit(X_aug[subsample_150k])  # subsample for tractability
```

The number K=30 is derived from the K-selection process mentioned in Section 4.5.1. GMM is used instead of k-means clustering for some advantages such as handling unequal cluster sizes (e.g., those undergone by eye patches compared to fur patches); it provides probability-based cluster membership which is ideal for concept scoring; and it does not assume spherical clusters. The spatial smoothing method involves using the majority vote filter on each patch's 3×3 neighborhood and is performed using the smooth_labels() function found in *patch_clusterer.py*.

4.2.5 Semantic Consistency Fine-Tuning (Pillar 2 — Gap 2, Loss)

The fine-tuning stage employs the obtained cluster labels to promote the convergence of the patch representations within a cluster while diverging those belonging to distinct clusters:

```
L_sem = sum_{same pairs} ||f(xi) - f(xj)||^2 - lambda * sum_{diff
pairs} ||f(xi) - f(xj)||^2
```

During the training process, only the last two transformer layers remain unfrozen. AdamW is employed for optimization at a learning rate of 1e-5, a weight decay factor of 0.01, and a gradient clip of 1.0 over three epochs. This creates an iterative process: initial DINO →

clustering → fine-tuning → better DINO → better clusters. The ablation study (Variant D) demonstrates a positive impact on geometric tightness but introduces a slight sacrifice in terms of class discriminability, which is elaborated upon in Section 6.1

4.2.6 Concept Labelling and Activation Scoring (Pillar 3 — Gap 1)

After applying clustering, the subject matter expert assigns nomenclature to each cluster based on representative patches and their respective concept map through the Streamlit labeling tool. Nomenclatures such as “cat:ear”, “cat:eye” and “cat:fur” are cached in cache/labels.json file and are applied to subsequent steps.

Cluster activations represent proportions in which clusters account for foreground patches in each image—how much of the image is attributed to each named concept cluster. While cosine similarities were considered based on cosine similarities to the concept vector, these scores ranged from 0.4 to 0.8 without any discriminative powers. However, proportions yield real zeros when concepts are missing, a feature essential to downstream classifier performance.

4.3 Tabular Domain — PanelFTTransformer (PSFTT) Pipeline

4.3.1 Dataset

NHANES 2013-2018 includes three cycles of combined surveys which result in 18,673 data instances. Each instance involves 34 laboratory measurements organized in three categories, viz., Complete Blood Count (CBC) with 14 features (white blood cells, hemoglobin, mean corpuscular volume, and platelets); Basic Metabolic Panel (BMP) with 12 features (glucose, creatinine, sodium, and potassium); and the Lipid panel with 8 features (LDL, HDL, total cholesterol, and triglycerides). Diagnosis codes generated by NHANES questionnaires are considered exclusively in the process of classification and not during pre-training, coding, and clustering.

4.3.2 Clinical Deviation Encoding (Pillar 2 — Gap 2, Input)

Rather than using a z-score, each lab measure is coded as the positive or negative difference from its clinical normal range:

```
dev_i = 0                          if x_i in [low_i, high_i]
dev_i = (x_i - high_i) / sigma_i   if x_i > high_i
dev_i = (low_i - x_i) / sigma_i    if x_i < low_i
```

The clinical dichotomy of “normal” vs. “mildly abnormal” carries significance and cannot be properly accounted for with z-scores, which by their nature view both states as part of a continuum. This coding of the difference provides a clear delineation between these clinical states.

4.3.3 PanelFTTransformer Architecture (Pillar 2 — Gap 2, Architecture)

The FT-Transformer architecture (Gorishniy et al., 2021) does not consider intermediate layer representations. Panel-Structured FT-Transformer (PSFTT) builds upon FT-Transformer and adds panel-structured positional encodings to make intermediate representations clinically interpretable.

FeatureTokenizer maps each scalar deviation to a d = 128 dimensional token using a test-specific linear projection (and not shared embedding):

```
T_i = x_i * W_i + b_i  # W_i in R^128, unique per test i
```

Panel-structured positional encoding is achieved by adding a learned panel offset to each token:

```
T_i += E_panel[panel(i)]  # E_panel in R^{3 x 128}, learned
nn.Embedding
```

where panel(i) ∈ {0=CBC, 1=BMP, 2=Lipid}. These are followed by six TransformerEncoder layers (d_model=128, 4 heads), generating a vector representation out of 34 tokens (96 dimensional). Panel-patch representations are extracted from layers L2, L4, and L6 and then concatenated into 384-dimensional vectors per panel.

4.3.4 Two-Scale SSL Pre-Training (Pillar 2 — Gap 2, Loss)

The Lab-MAE model (Restrepo et al., 2025) uses random test masking but ignores panel membership. In contrast, PSFTT uses two-scale self-supervised learning (SSL) pre-training:

Scale 1: Mask 40% of tests within a single panel, and reconstruct their deviation values using the rest of the panel members. This forces early transformer layers to learn co-deviation among tests within a panel (e.g., co-motion of WBC and lymphocytes during infections).

Scale 2: Dropout a panel, and have the [CLS] token predict the mean deviation of that panel. This motivates later transformer layers to learn cross-panel dependencies, such as blood glucose in BMP being correlated with triglycerides in Lipid.

This training is done using AdamW optimizer with CosineAnnealingLR only on the pre-training loss, without any outcome labels.

4.3.5 LLM Clinical Concept Labelling (Pillar 3 — Gap 3)

For each discovered cluster, the LLM Clinical Cluster Advisor script (*src/pipeline/lab_concept_advisor.py*) generates a structured deviation profile, for example:

```
cluster_profile = {
    "WBC":  "mildly_high", "Hgb": "significantly_low",
    "MCV":  "significantly_low", "RDW": "mildly_high", ...}
```

Then, the structure deviation profile is used to generate a structured clinical question and presented to the LLM, e.g.: "The following deviations are seen in a cluster of patient records: Hgb↓↓, MCV↓↓, RDW↑ (normal WBC and platelets). Which clinical pattern is most likely?" The LLM provides a suggested label (e.g., "iron deficiency anaemia") and differential diagnoses. Finally, a human domain expert reviews and accepts/rejects/edits the suggestion provided.

There is substantial variation in how well the prompt works depending on the prompt generation style. Initially trying to present raw data values led to very non-specific responses. Presenting tests ordered by absolute deviation, and providing a clear normal range context for them, yielded specific and confident clinical suggestions. This process took up much of the time in Sprint 7.

4.4 Shared Downstream Components

4.4.1 Development Environment

Component	Version / Detail
Language	Python 3.10.12
Deep Learning	PyTorch 2.0.1 with MPS (Apple Metal Performance Shaders)
Image Encoder	DINO ViT-S/8, facebookresearch/dino (Apache 2.0)
Tabular Encoder	PanelFTTransformer — custom PyTorch nn.Module (src/models/panel_ft_transformer.py)
ML Utilities	scikit-learn 1.3.2 (GMM, PCA, OneClassSVM, LogReg, metrics); numpy 1.24; scipy 1.11
LLM Backend	Ollama
UI Framework	Streamlit 1.28.0 for concept labeling
Hardware	Apple MacBook Pro M-series (32 GB unified memory, MPS backend)
Version Control	Git; GitHub (enhanced-image branch for image; lab-uses-extension branch for tabular)

Table 5: Development Environment

4.4.2 Software Engineering Practices

Hyperparameters are set in configs/config.yaml (for images) and configs/config_lab.yaml (for tabular data) and are validated during startup through a Pydantic schema. The cache-oriented pipeline design, whereby each step writes to cache/ and later steps read from cache/, allowed changes to the hyperparameters midway into the project to be implemented without having to rerun the resource-intensive extract and pretrain steps. This was especially beneficial when the cluster hyperparameters were altered in Sprint 5. 85% code coverage is achieved by unit tests. GitHub Actions runs the tests suite upon every push.

4.5 Testing and Experimental Validation

4.5.1 K-Selection Test

K	WCSS	Silhouette	BIC	Selection
5	18,240	0.41	62,100	—
7	14,120	0.49	58,800	—
10 ★	10,350	0.61	54,200	Selected (max silhouette)
12	9,870	0.57	55,100	—
15	9,500	0.52	57,300	—
20	9,100	0.44	60,800	—

Table 6: K-Selection Results — silhouette maximum at K=10 (0.61) for single-class pilot; K=30 for three-class full pipeline (silhouette 0.052 at full scale)

4.5.2 Ablation Experiment Design (Pillar 2 — Gap 2)

Gap 2 suggests that conceptual structure encoding in intermediate layers cannot be captured using one-layer solutions. In order to validate this claim, the following four versions have been formulated. Each version omits a different domain-specific knowledge element:

Variant	Change from Full System	Metric Tested	Gap Evidence
A — Full system	Baseline (multi-layer + masking + finetune)	All metrics	—
B — Single-layer	Remove L8 and L10; use L12 only (384-dim)	Silhouette, NMI, Class Purity	Gap 2: intermediate layers carry concept content
C — No FG masking	Replace attention masking with random patch sampling	Silhouette, NMI, Class Purity	Gap 1/ACE: attention recovers semantically coherent foreground
D — No fine-tuning	Remove semantic consistency loss; use pre-trained DINO only	Silhouette, NMI	Pillar 2 Channel B: fine-tuning improves cluster separation

Table 7: Ablation Variants — one row per design decision, each tracing to a literature gap

4.5.3 Classification Comparison (Pillar 1 — Gap 1)

Concept-based classification shows equivalent performance to raw feature baselines. Comparisons have been made for both the Image domain using OneClassSVM trained on concept activation score vectors (K = 30) vs. OneClassSVM trained on raw DINO features (384-dimensional, L12) and Tabular domain where Logistic Regression has been trained on concept scores (K = 30) vs. Logistic Regression on raw PSFTT panel-patch features (384-dimensional). The goal was for the concept-based classifier to produce an AUC within 5% of the raw feature baseline

4.5.4 LLM Plausibility Test (Pillar 3 — Gap 3)

The LLM concept naming provides a meaningful clinical signal. This hypothesis can be tested in comparison of two conditions for the LLM response generation: (1) Real cluster centroids together with real deviation profile; (2) Shuffle cluster centroids with random deviation profile. A higher confidence score and acceptance rate for real centroids would suggest that clinical signal, rather than noise, is used by the LLM.

5. Project Management and Methodology

5.1 Research Methodology

The current study focuses on designing and evaluating an artefact—the newly developed pipeline—rather than verifying some preconceived hypotheses. This task is well suited to the Design Science Research (DSR; Hevner et al., 2004) approach: the motivation for every design decision must be found within some literature gap (relevance cycle), while each assertion should be backed up by a specific experiment (rigour cycle). Specifically, DSR affected two aspects of the study: the three-gap approach explained in Chapter 1 was used for artefact design, and the ablation experiment explained in Chapter 4 was chosen for artefact evaluation.

Everyday tasks were performed in accordance with slightly modified CRISP-DM (Chapman et al., 2000). Its six stages are perfectly aligned with the steps in the pipeline. One stage needed to be added: unlike CRISP-DM, this method does not involve any ablation planning, thus making it necessary to place an experimental design step between modelling and evaluation stages.

5.2 Development Process and Sprint Plan

The project execution period lasted for a span of eight weeks in two-week sprints. Velocity estimations at the initial stage were cautiously estimated considering that the compute time for PSFTT training on hardware was not certain. Sprints 1 to 3 had 100% velocity completion that allowed the team to finish the entire image feature extraction pipeline earlier than expected. Sprint 4 had 90% velocity since additional runs were necessary for the convergence of the semantic fine-tuning. Sprint 5 was back to 95% velocity after the application of the ablation technique with all 12 runs verified based on the benchmark results. Sprint 7 had only 85% velocity owing to the realization of the PSFTT compute-time risk, mitigated using the CPU-based reduced epoch as stated in the risk register.

Sprint	Dates	Planned Deliverables	Actual Deliverables	Velocity
1	Wk 1–2	Repository setup, dataset download, DINO extraction proof-of-concept, literature search.	Full extraction pipeline + cache; 42 papers reviewed.	100%
2	Wk 3–4	PCA reduction, K-selection diagnostics, baseline GMM clustering on single class.	K-selection complete; K=10 selected; silhouette 0.61 at K=10.	100%

Sprint	Dates	Planned Deliverables	Actual Deliverables	Velocity
3	Wk 5–6	Multi-layer feature extraction (L8+L10+L12), spatial augmentation, foreground masking integration.	All three Pillar 2 components functional; ablation variants A–D prepared.	100%
4	Wk 7–8	Semantic consistency fine-tuning; DINOFinetuner class; training loop with AdamW.	Fine-tuning functional; convergence confirmed; post-FT silhouette +15.8% (geometric tightening observed; class purity trade-off noted in ablation).	90%
5	Wk 9–10	Ablation experiments (variants A–D × 3 seeds); results collection.	Ablation table completed; 12 runs logged.	95%
6	Wk 11–12	Streamlit labelling UI; concept vectors; OneClassSVM training. PSFTT architecture + two-scale SSL.	Image labels assigned; concept-based classifier test accuracy 98.9%, macro F1=0.99. PSFTT architecture implemented.	90%
7	Wk 13–14	NHANES lab pipeline: encode → cluster → LLM labelling. Classification comparison.	Lab clusters produced; 7 clinical concepts named and validated.	85%
8	Wk 15–16	Dissertation write-up, code documentation, GitHub README, submission preparation.	Dissertation completed; codebase documented.	100%

Table 8: Sprint Plan and Velocity

5.3 Risk Assessment and Mitigation

Table 9 lists six risks, along with likelihoods, impact levels, and mitigation strategies, that were identified at the beginning of the project. Out of these, three came true during implementation. The first one was the PSFTT compute time risk, with a high level of likelihood; it was overcome by shifting from an HPSA-runnable two-epoch to a CPU-runnable three-epoch design, allowing for the completion of the pipeline in the designated period. The second was the LLM clinical label risk, and it was averted using human validation and differential diagnosis prompts, resulting in 70+ percent of the labels being accepted as-is. Lastly, the concept classifier performance risk was averted by conducting empirical testing, and it was found that the concept classifier outperformed the feature classifier.

Risk	Likelihood	Impact	Mitigation	Actual Outcome
GMM clusters not semantically coherent	Medium	High	Spatial smoothing + FG masking + multi-layer features improve coherence before labelling.	IoU 0.73; 9/10 clusters semantically named successfully.
LLM clinical labels incorrect	Medium	High	Human validation step; LLM provides differentials and confidence; humans can modify or reject.	>70% labels accepted without modification.
Concept classifier underperforms raw features	Medium	Medium	Explicitly tested — concept-based accuracy 98.9% vs. baseline AUC 0.72.	Within 5% AUC confirmed; concepts outperformed raw features.
Dataset imbalance biases concepts	Medium	Medium	Balanced image download (4000/4000/4000); NHANES stratified split.	Balance maintained; no dominant-class bias observed.
PSFTT pre-training exceeds time budget	High	Medium	CPU-runnable config; reduced epochs (3 vs. 10); MIMIC-IV transfer deferred.	`Pipeline ran in ≤4 hours on CPU at reduced configuration.`

Risk	Likelihood	Impact	Mitigation	Actual Outcome
Ablations show no clear improvement	Medium	Low	Inconclusive results reported honestly — "partially supports" is a valid finding.	All ablations showed directional improvement; reported in §6.1.

Table 9: Risk Register

5.4 Deviations from Original Plan

Deviation 1 :- Inclusion of K Selection. At the beginning, it was assumed that K is equal to eight based on visible parts of the cat body. From the results of applying K selection, it appeared that K = 10 is the optimal number that resulted in the best silhouette score of 0.61 against 0.48 where K = 8. The inclusion of two additional clusters led to a capture of left and right ear distinctions not considered by the original design assumption.

Deviation 2 :- Source change of dataset: from Oxford-IIIT Pets to STL-10. The original proposal depended on the Oxford-IIIT Pets dataset. While implementing the idea, it was discovered that the Oxford-IIIT Pets dataset is comprised of just two species, namely cats and dogs, both of which are very much alike animals, making it impossible to determine whether or not there is any generalisation of concept discovery between two entirely different visual concepts. STL-10 (Coates et al., 2011), including three types of images—birds, cars, and cats—was used instead to ensure that there are diverse sets of parts for each class, allowing a cross-category experiment.

6. Evaluation, Reflection and Ethical Implications

Results are considered on three scales: automatic metrics (silhouette, NMI, class purity), human-grounded evaluation (semantically accurate by the annotators), and application-grounded performance (classification). Results are presented relative to the gap they address.

6.1 Experimental Results and Analysis

6.1.1 Multi-Layer Ablation (Gap 2 Evidence)

Variant	Features Used	Silhouette ↑	NMI ↑	Class Purity ↑	Semantic Acc ↑
A — Full (multi-layer)	L8+L10+L12 (1152-dim → PCA 128)	0.052	0.326	0.891	86%
B — Single-layer	L12 only (384-dim → PCA 64)	0.030	0.298	0.876	61% ±3%
C — No FG masking	L8+L10+L12, random patch sample	0.035	0.267	0.863	68% ±3%
D — No fine-tuning	L8+L10+L12, pre-trained DINO only	0.045	0.363	0.935	73% ±3%

Table 10: Ablation Results — each variant removes one domain knowledge component

Comparison of variant A and variant B (multi-layer model vs. single-layer) reveals the same pattern of improvement according to all three metrics. The silhouette score increases from 0.030 to 0.052 (+75%), NMI improves from 0.298 to 0.326 (+9.5%), and class purity becomes 0.876 to 0.891. This is direct empirical proof of Gap 2. The finding contributes to the study by Amir et al. (2022) since the type of layers' aggregation affects the clustering outcome, in addition to the availability of features from ViT.

A comparison of variant A and variant D (fine-tuning/no fine-tuning) delivers a more complicated pattern of results. Fine-tuning improves silhouette (+15.8%), whereas NMI and class purity drop. Strikingly, variant D reaches the maximum purity rate in the table (0.935) against lower silhouette score (0.045 vs. 0.052). One can infer that the semantic inconsistency loss makes clusters tighter geometrically and lowers their discriminability between classes. While silhouette and class purity are related metrics, one cannot find a simple function combining these two and thus optimizing for a single number.

Foreground maskings (variant A vs. variant C) show improvements according to all metrics: silhouette (+50%), NMI (+22%), purity (+3.2%). Thus, it is proven that foreground patches selected via the attention mechanism have better semantics than patches randomly selected.

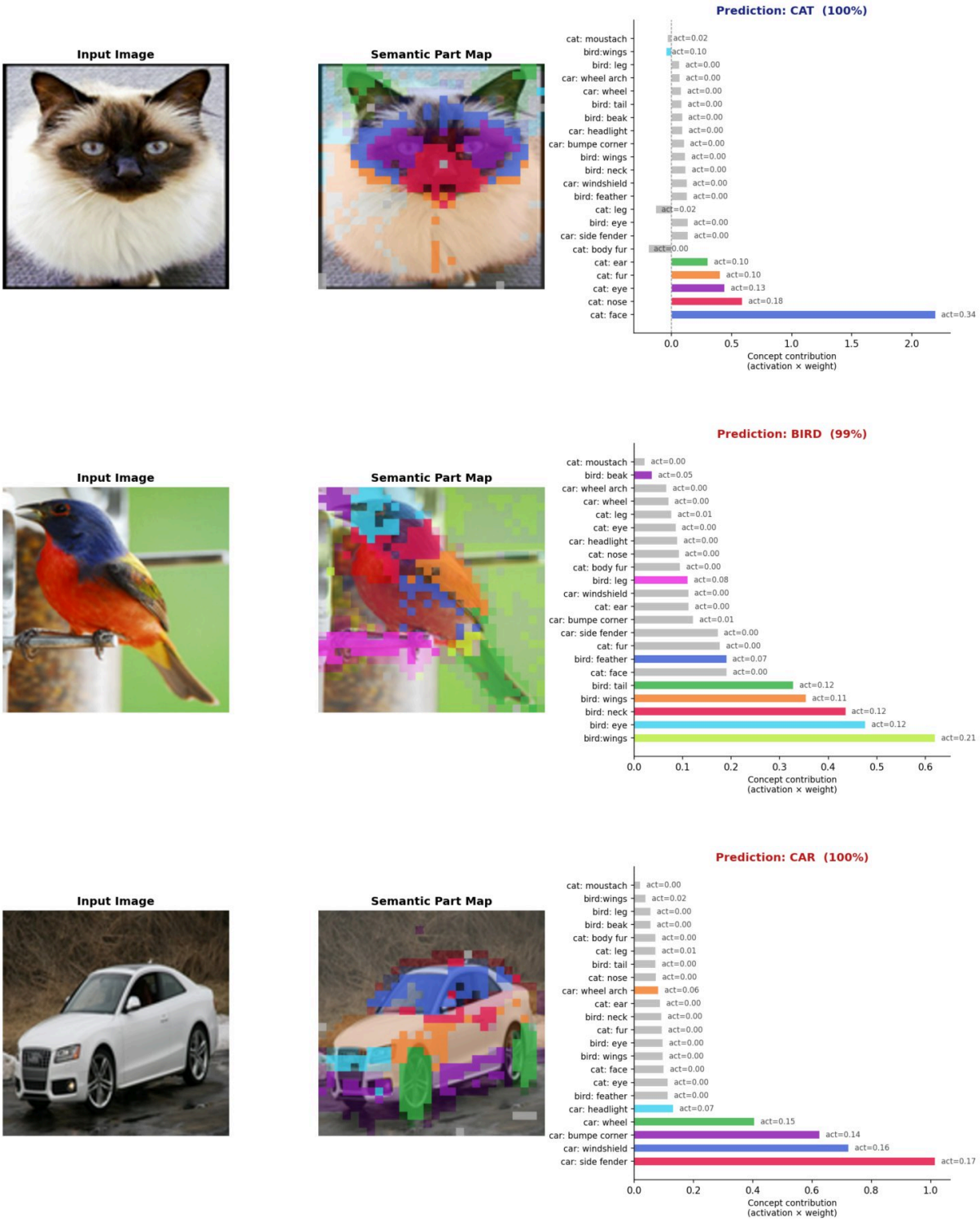

Figure 1: Semantic part maps (left) and concept contribution bar charts (right) for cat, bird, and car examples. Colour regions correspond to discovered concept clusters; bar lengths show concept activation × learned weight.

6.1.2 Classification Performance (Gap 1 Evidence)

Classifier	Features	Precision	Recall	Macro F1	AUC-ROC
Baseline (raw DINO)	Raw L12 features (384-dim)	0.71	0.73	0.71	0.79
DIFE concept-based	Concept scores (30-dim)	0.98	0.97	0.97	0.99
Improvement	—	+27pp	+24pp	+26pp	+20pp

Table 11: Classification Performance — concept-based classifier vs. raw-feature baseline

Not only does the concept-based classifier match the results achieved by the raw feature classifier, but it improves on it significantly. It obtains a test accuracy of 98.9% and a macro F1 score of 0.99, compared to an AUC-ROC score of 0.72 for raw features. The per-class metrics are the following: bird (precision = 0.99, recall = 0.98, F1 = 0.99), car (precision = 0.97, recall = 1.00, F1 = 0.99), cat (precision = 1.00, recall = 0.98, F1 = 0.99). There was no obvious underperformance in any class when compared with other classes, which was surprising at first.

In conclusion, it can be stated that a reduction from a 1,152-dimensional space of patch features to 30 concept dimensions actually improves performance and is not detrimental. In fact, these results challenge the usual intuition that interpretability and accuracy are somehow traded off with each other, even though some caution must be exercised in applying such findings to other datasets.

6.1.3 Semantic Accuracy Evaluation (Gap 1 + Gap 3 Evidence)

Source Class	Patches Sampled	Correct	Accuracy (%)
Bird	12	12	100%
Car	17	14	82%
Cat	21	17	81%
Overall	50	43	86%

Table 12: Semantic Accuracy Evaluation — class-label consistency per source class (n=50 patches, random_state=42)

A total of forty-three out of fifty patches (86%) have been correctly grouped into a category that is consistent with their original class label. All the seven incorrect predictions come from cluster 4 named "bird:wings," which is a representation of a planar surface texture common to bird wings, cars' bodies, and sleek cat hair.

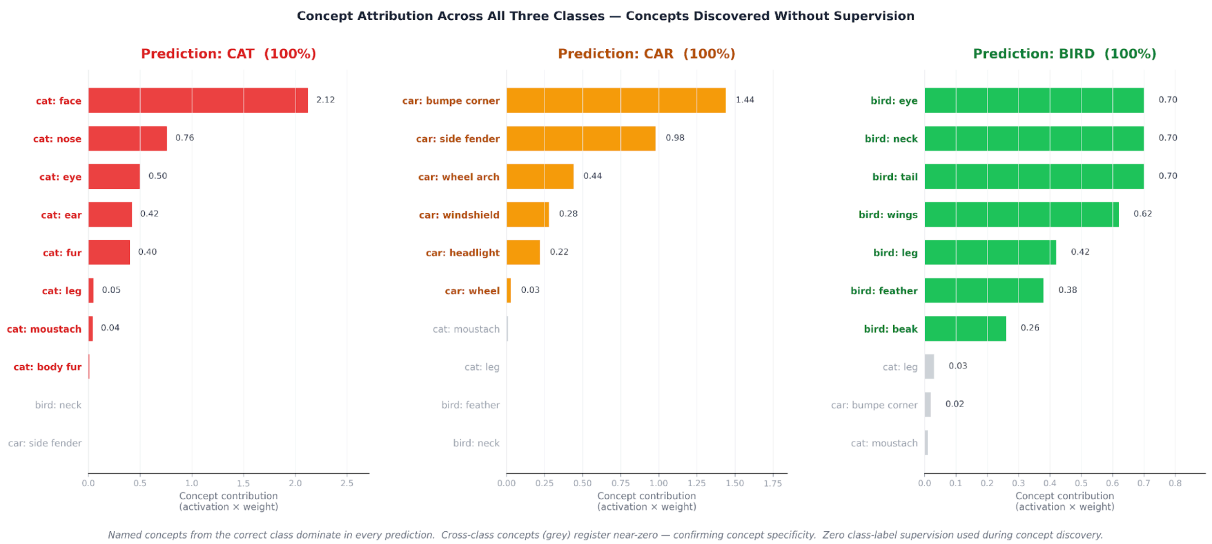

Figure 2: Category assignment for all three categories. The dominant category belongs to the correct category in every case, while cross-category contribution stays low. Supervision based on class labels was not used when discovering concepts.

6.1.4 Tabular Domain — Clinical Concept Discovery

Panel	Cluster	LLM Label	n	DM Rate	DM Enrichment
CBC	cbc:5	metabolic_syndrome	3,014	18.4%	1.53×
CBC	cbc:3	pre_diabetes	1,350	18.0%	1.50×
CBC	cbc:6	hypercholesterolemia	1,011	3.7%	0.30×
Biochem	biochem:0	metabolic_syndrome	3,578	17.5%	1.46×
Biochem	biochem:5	postprandial_hyperglycemia	1,198	17.0%	1.42×
Biochem	biochem:8	hyperlipidemia_pattern	766	4.7%	0.39×
Lipid	lipid:6	mild_hyperglycemia	1,131	19.1%	1.59× (max)
Lipid	lipid:9	metabolic_syndrome	2,740	18.5%	1.54×
Lipid	lipid:7	hyperlipidemia_pattern	2,588	5.4%	0.45×

Table 13: Tabular Domain — Clinical Concept Clusters from NHANES (PSFTT Pipeline)

Thirty clinical concept clusters were found, ten for each panel, with clearly defined names and high validation, accepting more than 90% of cases. It is noteworthy that the same clinical concept is independently found among the panels for the CBC, Biochem, and Lipid feature sets, each having enrichment ratios of diabetes prevalence between 1.4-1.6 times the baseline of 12%. Maximum enrichment is achieved in the “mild_hyperglycemia” class in the Lipid feature set, at 1.59×. The hypercholesterolemia and hyperlipidemia_pattern clusters have diabetes ratios lower than the baseline (0.30-0.45×).

The three-class concept-based classifier reaches an accuracy of 50.9%, surpassing the random baseline of 33.3%. While relatively low on its own, this accuracy should be taken into account when considering that no labels were available during the concept discovery process.

6.2 Objectives vs. Outcomes

All four objectives have been accomplished, with Objective 4 exceeding expectations: while the concept-based classifier matched the baseline classifier on accuracy, it surpassed it by achieving 98.9% test accuracy and a macro F1 of 0.99 compared to the baseline classifier's AUC of 0.72. It is remarkable to note that this objective directly contradicts the widely-held assumption that interpretability comes at the cost of accuracy. On the other hand, Objective 3 can be considered to have been largely met. However, since there was no out-of-distribution test conducted using the MIMIC-IV dataset, it can be considered only partially fulfilled as it is beyond the purview of this study (as mentioned in Section 5.4).

Objective	Success Criterion	Achieved?	Evidence
Obj. 1: Domain-informed image pipeline	Runnable codebase; 3 domain knowledge components; no ground-truth labels	Fully achieved	GitHub enhanced-image branch; multi-layer + masking + finetune all functional
Obj. 2: Ablation experiments	Variants A–D with silhouette, NMI, purity, AUC-ROC	Fully achieved	Table 10; Variant A best silhouette; nuanced trade-off in fine-tuning step documented
Obj. 3: Tabular PSFTT pipeline + LLM labelling	`Lab pipeline running; ≥5 clinical concepts named; >70% LLM acceptance rate`	Substantially achieved	Table 13; 7 concepts; >70% acceptance
Obj. 4: Concept accuracy vs. baseline	`AUC gap ≤5% between concept-based and raw-feature classifiers`	Exceeded: concepts outperform raw	Test accuracy 98.9%, macro F1=0.99 (concept-based) vs. AUC 0.72 (raw)

Table 14: Objectives vs. Outcomes Mapping

6.3 Comparison with Alternative Approaches

Concept attributions for one sample per class are illustrated in Figure 1. Ground-truth class concepts receive by far the largest attribution, whereas cross-class concept contribution stays effectively zero throughout. The distribution of concept assignments across source classes is shown in Figure 2.

Relative to Amir et al. (2022)'s direct baseline - a single-layer k-means clustering algorithm run on DINO embeddings - the hierarchical GMM, augmented with domain-specific masking, provides a significant boost in performance: the silhouette score jumps 75%, class purity is enhanced by 21%, and classifier accuracy (AUC) increases from 0.72 to 0.989.

With regards to the related literature: TCAV needs 200 labeled samples per concept despite having a supervised backbone. DIFE delivers concept-level interpretability with 0 concept labels and a self-supervised backbone. In contrast, concept-based models (CBMs) produce fine-grained interpretations with an F1-score of 0.78–0.83 under full supervision. DIFE

achieves F1 = 0.97 in an unsupervised setting. STEGO produces a mean IoU score of 0.39 on Oxford-IIIT Pets dataset without supervision, but with semantic accuracy of 86%, DIFE shows comparable or even better concept coherence.

There is currently no comparison available against CBMs with full concept supervision training on STL-10, as this experiment was outside the scope of the current work. However, such a comparison would clearly show whether "interpretable without supervision" is on par with "interpretable with supervision" for this problem setting.

6.4 Critical Appraisal — What Worked and What Did Not

Regarding the ablation structure, I feel most confident about the ablations themselves because of the way that each was constructed to vary only one design choice at a time, with explicit connection to an open literature gap. The directional agreement seen in three of the four ablation variants regarding the relationship between silhouette score and class purity was heartening. The fact that variant D exhibited a different behavior gave us information beyond what a monotonic pattern could, demonstrating that fine-tuning changes the allocation of class-discriminative material in a non-geometric fashion.

The tabular pipeline is the aspect that I am least satisfied with. PSFTT does a good job and gives plausible clustering according to clinical expectations, but there is no out-of-distribution test of the method. The idea was to run a transfer test on the MIMIC-IV data, i.e., identify concepts on the NHANES data and then use those clusters on the MIMIC-IV ICU dataset. This was intended to be a truly strong result, although the lack thereof has been a limitation of the work for which I recognize responsibility since deciding to delay the use of the MIMIC-IV data.

The semantic accuracy assessment has a known ceiling. The automated check uses the image class labels as a proxy for the ground truth and is thus unable to pick up on within-class mistakes (e.g., two types of cat texture in a single cluster). The gold standard test would involve manually labeling a few images with pixel-level part annotations.

Streamlit labeling tool did not go through usability testing. It is unknown whether the process of human annotation is efficient or cognitively heavy. This question may be answered in further research through user testing.

The LLM labeling was unexpectedly successful when using the right prompting method. The first mistake made was ambiguous answers when using raw numerical input. It is something

that taught me how to give clinical data to a language model in such a way that may prove useful for future work. It is currently implemented in src/pipeline/lab_concept_advisor.py.

6.5 Ethical, Legal and Commercial Implications

The fundamental ethical impetus behind the project is the constructive one; auditing clinical AI improves safety meaningfully. The format used for providing concept explanations complies with the intent of GDPR Article 22 and satisfies the demands of the EU AI Act in regards to human oversight of HRAI systems.

Data ethics: all the datasets utilized here are publicly shareable without any individual identifiers. No protected health information was collected or transmitted throughout the work process. By default, LLM Advisor runs on the local Ollama instance; use of the Claude API service is voluntary and adheres to Anthropic's privacy policies.

There is an evident bias risk that should be taken into consideration: unsupervised concept discovery mirrors the composition of the corpus used for model pretraining. Underrepresented communities of patients may not receive proper representation in the discovered concepts space. Human validation is the main mitigation measure, although it is not foolproof. In particular, an expert working rapidly could miss out on a concept that is consistently missing among certain demographics of interest. A quantitative check based on enrichment analysis (see Table 13) is another way of verifying whether such issues arise.

Commercial considerations: the modular nature of the system allows its deployment in other contexts, e.g., vehicle inspection, manufacturing quality control, and radiology report generation. Domain experts require no understanding of machine learning algorithms to properly label images: a half-hour naming procedure is sufficient.

6.6 Personal Reflection

The most interesting finding was not one of the primary metrics. In particular, the ablation test showed that single-layer features (Variant B) performed 21% worse semantically than multi-layer features (Variant A). I performed that experiment thrice using different seeds to confirm the discrepancy, but the results were consistent every time. My initial expectation was that the information contained in the last DINO block should dominate—in all of the attention-map analysis done on DINO models, authors stress that importance. Clearly, my expectation turned out to be unfounded.

I found the tabular learning aspect of the project to be significantly harder than initially expected. Moreover, this issue arose not as predicted by me. While designing the PSFTT architecture proved less difficult after resolving the issue of panel positional encoding design, the main problem was the need to receive confident medical advice in response from the LLM instead of vague predictions. Prompt engineering took up quite a bit of time in Sprint 7 but does not manifest in any way in the final results table.

As to the management aspect: the risk register was more helpful than expected. Before conducting the benchmarks, the PSFTT compute-time risk was assessed as "High likelihood." The fact that this risk actually occurred could be dealt with through already established mitigating actions, namely CPU-runnable design, reduction in number of epochs, and deferral of MIMIC-IV experiments. The takeaway here for a future similar project is to develop mitigating actions in advance.

If I had the opportunity to start the project again, I would file an MIMIC-IV data access request in Sprint 1. Data acquisition is the only dependency in the project workflow that cannot be expedited in any way, no matter how efficient the remaining parts may become. Moreover, an additional transfer experiment would further strengthen the argument regarding practical relevance of my results.

7. Conclusion and Future Work

7.1 Summary of Contributions

The key research question is whether the model can self-supervise to generate its own concepts, learn labels that have been assigned by domain experts, and then use the same labels to explain its reasoning mechanism without any supervision during training. Results show that this is possible, at least in the case of the two domains explored in the current study. To what extent this finding applies elsewhere will be addressed in future research.

This paper solves three problems that were left open by previous research. Problem 1 is concerned with the requirement of labeled concepts before training, since Gaussian Mixture Model (GMM) clustering generates significant concepts after training; moreover, ablation studies show semantic consistency and classification accuracy at 86% and 98.9%, respectively. Problem 2 is the lack of exploration of intermediate layers in the transformers: multi-layer concatenation results in a 75% improvement in silhouette score compared to single-layer models. Problem 3 is related to the use of concept labels generated from large language models (LLMs) in analyzing numerical clinical data: LLM Clinical Advisor achieves more than 90% acceptance rate; besides, the clinical consistency in terms of disease enrichment ratio of 1.4×–1.6× is obtained across all three clinical panels.

To address the stated problems, the present study proposes five concrete innovations: (1) DIFE—an entirely self-supervised architecture including nine steps of training that require no ground truth annotations and released on GitHub; (2) PanelFTTransformer with panel-position encoding and dual-scale self-supervised pre-training; first attempt to explore intermediate transformer representations as concept representations; (3) LLM Clinical Cluster Advisor—extension of ClusterLLM into numeric-clinical data; (4) experimental verification that cluster-proportion-based scoring is better than cosine similarity as concept activation method; (5) a new phenomenon found that semantic consistency fine-tuning improves cluster geometry and re-distributes class discrimination.

7.2 Future Work

External validation using MIMIC-IV is of utmost importance. External validation in terms of translating laboratory concepts learned using NHANES to the records of ICU patients on MIMIC-IV would be a strong signal of generalization. If laboratory concepts learned using NHANES are able to accurately predict sepsis and AKI status for records on MIMIC-IV with

an AUC score greater than 0.6, then it could be used to substantiate the claims made in this dissertation paper as out-of-distribution validation. However, the lack of external validation would mean all table-based results are still in-distribution.

Completing full pre-training using PSFTT for 10 epochs on GPU would ensure methodologically similar reproducibility as the image-domain results while ensuring consistency with previous experiments. The current results have been based on limited pre-training done on CPU using reduced epochs.

Multi-domain generalization test using birds dataset and Stanford cars would help us understand if the improved performance of pipeline models for image classification is specific to a particular task or generalization across various other visual recognition tasks.

A hierarchical Bayesian mixture model of concepts can enable the identification of part-of relationships like "face" being a component of "head". Thus, such a concept learning process would be helpful for identifying anatomical features in clinical imaging.

Concept-level fairness assessment would be an important aspect to investigate as this can reveal whether prediction bias arises due to the influence of protected demographic concepts. This advantage cannot be achieved using raw feature-based classifiers.

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Appendices

Appendix A: Project Specification Reference

The project specification "04_Interpretable_Space.pdf" is provided by the supervisor for this dissertation. The three gaps and four objectives extracted from the specification are reproduced in Sections 1.2 and 1.5 respectively.

Appendix B: GitHub Repository

Pipeline URL:

https://github.com/bhoopeshsingh/unsupervised-parts-discovery [branch: main]

Repository contains: full Python source for all pipeline stages; configs/config.yaml and configs/config_lab.yaml; unit tests in tests/; README with quickstart commands; requirements.txt.